Plan

- Basic Feasible Solution (BFS)
- Non degenerate BFS
- Degenerate BFS
- Simplex Algorithm

- Consider LPP **(P)**
  Max or Min $\mathbf{c}^T\mathbf{x}$
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- The variables $x_1, \ldots, x_m$ are called **basic variables**, and $x_{m+1} = x_{m+2} = \ldots = x_n = 0$, are called **non basic variables** of the **BFS** $\mathbf{x}$

Degenerate BFS

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- If LPP **(P)** is the diet problem, (with $\geq$ inequalities changed to equalities in the original problem),
  then $\mathbf{x}$ gives the quantities of the food products F_j, $j = 1, \ldots, n$ in the diet

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$$\tilde{\mathbf{a}}_k = \sum_{i=1}^{m} u_{ik}\tilde{\mathbf{a}}_i = [\tilde{\mathbf{a}}_1 \ldots \tilde{\mathbf{a}}_m] \begin{bmatrix} u_{1k} \\ \vdots \\ u_{mk} \end{bmatrix} \qquad \textbf{(*)}$$

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- Then $\begin{bmatrix} u_{1k} \\ \vdots \\ u_{mk} \end{bmatrix} = B^{-1} \tilde{\mathbf{a}}_k$ for all $k = 1, 2, \ldots, n$

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- For all $k = 1, \ldots, m$, $z_k = c_k$
- If $z_k > c_k$ then look for a better solution

- the simplex table corresponding to BFS $\mathbf{x}$ is given by

	0	..	0	..	$c_s - z_s$	..	$c_k - z_k$	..	
	$B^{-1}\tilde{\mathbf{a}}_1$	..	$B^{-1}\tilde{\mathbf{a}}_m$	..	$B^{-1}\tilde{\mathbf{a}}_s$	..	$B^{-1}\tilde{\mathbf{a}}_k$	..	$B^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_1$	1	..	0	..	u_{1s}	..	u_{1k}	..	x_1
$\tilde{\mathbf{a}}_2$	0	..	0	..	u_{2s}	..	u_{2k}	..	x_2
$\vdots$	0	..	0	..	$\vdots$	..	$\vdots$	..	$\vdots$
$\tilde{\mathbf{a}}_r$	$\vdots$	..	$\vdots$	..	u_{rs}	..	u_{rk}	..	x_r
$\vdots$	$\vdots$	..	$\vdots$	..	$\vdots$	..	$\vdots$	..	$\vdots$
$\tilde{\mathbf{a}}_m$	0	..	1	..	u_{ms}	..	u_{mk}	..	x_m

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- If $u_{is} > 0$ for some $i = 1, \ldots, m$ then $x'_i \geq 0$ only if $x'_s \geq 0$, is such that $x'_s \leq \frac{x_i}{u_{is}}$

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- $\frac{x_r}{u_{rs}}$ is called the **minimum ratio**
- $\mathbf{x}' \geq \mathbf{0} \Rightarrow x'_s \leq \frac{x_r}{u_{rs}}$
- Also $A\mathbf{x}' = \mathbf{b}$ which implies $\mathbf{x}' \in Fea(LPP)$

- $\mathbf{c}^T\mathbf{x}' = \sum_{i=1}^{m} c_i x'_i + c_s x'_s$

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  $= \mathbf{c}^T\mathbf{x} + x_s'(c_s - z_s)$

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- x_s is called the **entering variable**, and x_r is called a **leaving variable**

- $\mathbf{c}^T\mathbf{x}' = \sum_{i=1}^{m} c_i x'_i + c_s x'_s$
  $= \mathbf{c}^T\mathbf{x} + x'_s(c_s - z_s)$
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- $\mathbf{c}^T\mathbf{x}' < \mathbf{c}^T\mathbf{x}$ if $x'_s > 0$ which is when the **minimum ratio** $\frac{x_r}{u_{rs}} > 0$
- Let $x'_s = \frac{x_r}{u_{rs}}$
- Then $x'_r = x_r - u_{rs}\frac{x_r}{u_{rs}} = 0$
- x_s is called the **entering variable**, and x_r is called a **leaving variable**
- If there exists $r, t \in \{m+1, \ldots, n\}$
  $\frac{x_r}{u_{rs}} = \frac{x_t}{u_{ts}} = min\{\frac{x_i}{u_{is}} : u_{is} > 0\}$
  then take any **one** of r, t as the **leaving variable**

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- $\mathbf{b} = \sum_{i=1, i\neq r}^{m}(x_i - \frac{u_{is}}{u_{rs}}x_r)\tilde{\mathbf{a}}_i + \tilde{\mathbf{a}}_s(\frac{x_r}{u_{rs}})$

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- If z'_k denotes the new values of z_k then
  $z'_k = z_k + \frac{(c_s - z_s)}{u_{rs}} u_{rk}$
  and $c_k - z'_k = (c_k - z_k) - \frac{(c_s - z_s)}{u_{rs}} u_{rk}$

- The simplex table corresponding to the new BFS $\mathbf{x}'$ is given by

	$c_s - z'_s$ $c_s - z_s - \frac{(c_s - z_s)}{u_{rs}} u_{rs} = 0$	$c_k - z'_k$ $(c_k - z_k) - \frac{(c_s - z_s)}{u_{rs}} u_{rk}$	
	$B'^{-1}\tilde{\mathbf{a}}_s$	$B'^{-1}\tilde{\mathbf{a}}_k$	$B'^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_1$	$u_{1s} - \frac{u_{1s}}{u_{rs}} u_{rs} = 0$	$u_{1k} - \frac{u_{1s}}{u_{rs}} u_{rk}$	$x_1 - \frac{u_{1s}}{u_{rs}} x_r$
$\tilde{\mathbf{a}}_2$	$u_{2s} - \frac{u_{2s}}{u_{rs}} u_{rs} = 0$	$u_{2k} - \frac{u_{2s}}{u_{rs}} u_{rk}$	$x_2 - \frac{u_{2s}}{u_{rs}} x_r$
$\vdots$	0	$\vdots$	$\vdots$
$\tilde{\mathbf{a}}_s$	$\frac{u_{rs}}{u_{rs}} = 1$	$\frac{u_{rk}}{u_{rs}}$	$\frac{x_r}{u_{rs}}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\tilde{\mathbf{a}}_m$	$u_{ms} - \frac{u_{ms}}{u_{rs}} u_{rs} = 0$	$u_{mk} - \frac{u_{ms}}{u_{rs}} u_{rk}$	$x_m - \frac{u_{ms}}{u_{rs}} x_r$

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	$c_s - z'_s$	$c_k - z'_k$	
	$c_s - z_s - \frac{(c_s - z_s)}{u_{rs}} u_{rs} = 0$	$(c_k - z_k) - \frac{(c_s - z_s)}{u_{rs}} u_{rk}$	
	$B'^{-1}\tilde{\mathbf{a}}_s$	$B'^{-1}\tilde{\mathbf{a}}_k$	$B'^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_1$	$u_{1s} - \frac{u_{1s}}{u_{rs}} u_{rs} = 0$	$u_{1k} - \frac{u_{1s}}{u_{rs}} u_{rk}$	$x_1 - \frac{u_{1s}}{u_{rs}} x_r$
$\tilde{\mathbf{a}}_2$	$u_{2s} - \frac{u_{2s}}{u_{rs}} u_{rs} = 0$	$u_{2k} - \frac{u_{2s}}{u_{rs}} u_{rk}$	$x_2 - \frac{u_{2s}}{u_{rs}} x_r$
$\vdots$	0	$\vdots$	$\vdots$
$\tilde{\mathbf{a}}_s$	$\frac{u_{rs}}{u_{rs}} = 1$	$\frac{u_{rk}}{u_{rs}}$	$\frac{x_r}{u_{rs}}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\tilde{\mathbf{a}}_m$	$u_{ms} - \frac{u_{ms}}{u_{rs}} u_{rs} = 0$	$u_{mk} - \frac{u_{ms}}{u_{rs}} u_{rk}$	$x_m - \frac{u_{ms}}{u_{rs}} x_r$

- The entry u_{rs} of the previous table which is made 1 (by dividing) in this table is called the **pivot element**

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  So given any $M \in \mathbb{R}$, by taking $x'_s > 0$ sufficiently large we can make $\mathbf{c}^T\mathbf{x}'$ smaller than M
  So the LPP has **NO** optimal solution

- $D = \{\mathbf{d} \in \mathbb{R}^n : \mathbf{d} \neq 0, A_{m \times n}\mathbf{d} = \mathbf{0}, \mathbf{d} \geq \mathbf{0}\}$ is the set of all directions of $S = Fea(LPP) = \{\mathbf{x} \in \mathbb{R}^n : A_{m \times n}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}, rank(A) = m\}$

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- If for some basis matrix B and a column $\tilde{\mathbf{a}}_s$ of A, $B^{-1}\tilde{\mathbf{a}}_s \leq \mathbf{0}$

$$\mathbf{d} = \begin{bmatrix} -B^{-1}\tilde{\mathbf{a}}_s \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

is an extreme direction of S,

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is an extreme direction of S, where the entry **1** in the above vector is at the s th position
- If $\mathbf{d}$ is as above then $\mathbf{c}^T\mathbf{d} < 0$

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- The above optimality conditions (for max and min problems) are sufficient but not necessary for an optimal solution

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  The simplex table corresponding to the **BFS** with basic variables x_2 and s_1 is given by:
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$c_j - z_j$	2	0	0	-1	
	x_1	x_2	s_1	s_2	$B^{-1}\mathbf{b}$
x_2	-1	1	0	1	1
s_1	1	0	1	-2	2

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$c_j - z_j$	1	1	0	0	
	x_1	x_2	s_1	s_2	$B^{-1}\mathbf{b}$
s_2	-1	1	0	1	1
s_1	-1	2	1	0	4

  Note that the above table is optimal

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$c_j - z_j$	0	0	1	0	
	x_1	x_2	s_1	s_2	$B^{-1}\mathbf{b}$
x_2	0	1	1	-1	3
x_1	1	0	1	-2	2

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x_2	-1	1	0	1	0	1
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	x_1	x_2	s_1	s_2	s_2	$B^{-1}\mathbf{b}$
s_2			0	1	0	4
s_1			1	0	0	1
s_3			0	0	1	9

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